



Exercise 1

Preprocessing & Regression

In this exercise, you will perform the initial step of every project — data preprocessing. You must prepare the dataset for your model and then train the most suitable type of regression on it.

Make sure to visualize and analyze your data thoroughly:

- Plot every relevant variable relationship (e.g., salary vs. experience, salary vs. education).
- Plot the full correlation matrix of numerical features.
- Interpret what these plots show about your data.

How to interpret the plots:

- **Variable relationship plots** (scatter, bar, or box plots) show how each feature relates to the target variable. For example, a positive trend in a scatter plot indicates that as one variable increases, the target tends to increase as well.
- The **correlation matrix** shows how strongly pairs of numerical variables are linearly related. Values close to +1 mean strong positive correlation, values near -1 mean strong negative correlation, and values around 0 indicate weak or no linear relationship.
- Use these insights to decide which features are most informative or redundant for your regression model.

Finally, upload your Jupyter Notebook (.ipynb) file to the course site. Good luck!

DATASETS:

- [Stack Overflow Annual Developer Survey 2024](#)
- [Stack Overflow Annual Developer Survey 2023](#)
- [Stack Overflow Annual Developer Survey 2022](#)
- [Stack Overflow Annual Developer Survey 2021](#)

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